

Problem

Design a problem to help other students better understand how to find the Laplace transform of different time varying functions.

Step-by-step solution

Step 1 of 3

Consider a time domain signal $f(t)$.

The Laplace transform of the signal is used to transform the time domain signal into complex frequency domain signal.

Depends on the range of t value, the Laplace transforms are classified into 2 types. They are unilateral transforms and bilateral transforms.

If the signal is defined for only $t > 0$, then the transform is known as unilateral transform.

$$L[f(t)] = \int_0^{\infty} f(t)e^{-st} dt \dots\dots (1)$$

If the signal is defined for $-\infty < t < \infty$, then the transform is known as bilateral transform.

$$L[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-st} dt \dots\dots (2)$$

The Laplace transform of signal $f(t)$ is defined only if the integration of the signal is convergent over a time period.

Step 2 of 3

Consider the following steps to determine the Laplace transform of time varying function.

First determine that the given signal is defined for only $t > 0$ or for $-\infty < t < \infty$.

If it is defined for $t > 0$, use equation (1) to determine the Laplace transform.

If it is defined for $-\infty < t < \infty$, use equation (2) to determine the Laplace transform.

Step 3 of 3

After finding the Laplace transform of $f(t)$, define the region of convergence for which the Laplace transform is existed.

Consider $s = \sigma + j\omega$.

$$L[f(t)] = \int_0^{\infty} f(t)e^{-(\sigma + j\omega)t} dt$$

Consider σ_c is some real value.

The region of convergence for the Laplace transform is $\text{Re}(s) = \sigma > \sigma_c$.

The Laplace transform is defined only within the region. It is not defined for outside the region.